MANAN SHAH

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SUMMARY

Software Engineer pursuing an MS in Computer Science with over a year of experience engineering backend microservices, cloud infrastructure, and mobile applications. Specialized in accelerating data processing pipelines, reducing system latency, and automating CI/CD workflows.

EDUCATION

Indiana University Bloomington, IN, USA **Aug 2024 – May 2026**
MS in Computer Science | **GPA: 3.81/4.00**

NMIMS University, Mumbai, India **Jul 2020 – May 2024**
B.Tech in Artificial Intelligence | **GPA: 3.41/4.00**

EXPERIENCE

Research Assistant | NMIMS University **Dec 2023 – May 2024**
Spearheaded the architectural development of a multi-stage AI pipeline and parallelized data system to generate 3D robotics simulations, accelerating experimental iteration cycles and reducing rendering artifacts to support internal autonomous systems research workflows.

- Architected **Apache Kafka** & **Python** data pipelines, resolving system I/O bottlenecks to cut model convergence from **6.5 to 2.5 hours**.
- Optimized multi-view **OpenCV** camera pose alignment algorithms, reducing the simulated visual rendering artifacts by exactly **28%**.
- Standardized **200+** reproducible testing environments by managing modular **YAML** configurations and **Git** version control branches.

Machine Learning Engineer Intern | STAT Institute **May 2023 – Oct 2023**
Expanded backend cloud infrastructure and microservices to support quantitative research workflows, integrating predictive time-series models into decentralized edge devices to improve high-frequency trading simulation accuracy across distributed computing environments.

- Launched **Node.js**, **REST**, and **GraphQL** microservices utilizing **Redis** caching, slashing network payload latency to exactly **195ms**.
- Configured **AWS EC2** Auto Scaling and tuned **CloudWatch** metrics thresholds, reducing the volume of system false alerts by **19%**.
- Benchmarked **50K+** concurrent **NumPy** batch processes on historical financial data to maximize the core data-processing throughput.

Software Engineer Intern | GNVS Institute **Dec 2022 – May 2023**
Developed core full-stack features for a cross-platform mobile career-networking application connecting students and recruiters, integrating real-time database synchronization and deploying CI/CD release pipelines to maintain system uptime during production releases.

- Shipped native **Android** (**Java**/**Kotlin**) and cross-platform **Flutter** application features across **8** active **Agile** Scrum sprints.
- Integrated **Firebase** Authentication workflows and **NoSQL** database architectures to enable real-time user state synchronization.
- Automated a **Jenkins** CI/CD pipeline implementation, writing unit tests to deploy **12** major releases with zero production bugs.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Java, SQL, C++, Kotlin, Dart
Backend & Database	Node.js, FastAPI, Express, REST/GraphQL/gRPC, MongoDB, PostgreSQL, MySQL, Redis, Apache Kafka, Django
Cloud & DevOps	AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Linux/Bash, Terraform
Front-end & Mobile	React.js, Flutter, Tailwind CSS, Android Studio
Dev Practices	Agile/Scrum, Git, Unit/Integration Testing, TDD, System Design, Microservices Architecture

PROJECTS

ChefPalAI: Your AI-Powered Kitchen Companion **Apr 2025**
Engineered a culinary web platform utilizing a React and Node.js microservices architecture to provide personalized dietary recommendations, structuring indexed NoSQL data models to guarantee sub-second retrieval times for complex nutritional queries and user preference routing.

- Programmed a full-stack **React**, **TypeScript**, and Node.js web application to deliver a responsive client front-end interface.
- Designed **MongoDB** compound indexes (ESR rules) combined with a **Redis** caching layer to guarantee strict **<1s** database lookups.
- Implemented data analytics dashboards utilizing **Matplotlib** and WebSockets to enable real-time tracking of user dietary visualization.

StockPulse: A Sentiment-Based Stock Forecasting Tool **Jun 2023 - Nov 2023**
Constructed an event-driven sentiment analysis engine utilizing FastAPI and Kafka to process high-throughput social media streams, powering financial forecasting dashboards to identify rolling market sentiment shifts for algorithmic trading and quantitative decision logic systems.

- Scaled a **FastAPI** backend via asynchronous request handlers to manage **500+** concurrent API connections per minute under load.
- Formulated an **Apache Kafka** and **Redis** event-driven engine to ingest live streaming data while maintaining **sub-50ms** latency.
- Synchronized backend **SQL** database pipelines to automatically refresh the data visualization dashboard metrics exactly every **5** minutes.

LIDNeRF: Language guided NeRF Editing with InstructDiffusion **Dec 2023 - May 2024**
Pioneered an ML engineering pipeline to streamline language-guided 3D object retexturing via diffusion models, orchestrating containerized GPU deployments to standardize hyperparameter tracking and accelerate compute-heavy rendering jobs across distributed computing clusters.

- Orchestrated containerized GPU parallel computing pipelines utilizing both **Docker** and **Kubernetes** orchestration frameworks.
- Accelerated automated **Python** multiprocessing deployment scripts, reducing the pipeline execution time from **9 to 3 hours** per run.